MULTIPLE IMPUTATIONS ON MISSING DATA IN ddRAD-SEQ PHYLOGENETICS OF A SMALL SAMPLING

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INTRODUCTION

Double digest restriction site-associated DNA sequencing (ddRADseq) offers a flexible and highly cost-effective strategy for plant genome sampling.

However, the method relies on the conservation of enzyme recognition sites to isolate homologous DNA fragments for sequencing, with the consequence that mutations disrupting these sites lead to missing information. Furthermore, the missing data is also born from sequenced samples that differed in the generated reads and coverage. This discrepancy was overcome by applying Multiple Imputations (MI) through different filtering set to the samples and called SNPs.

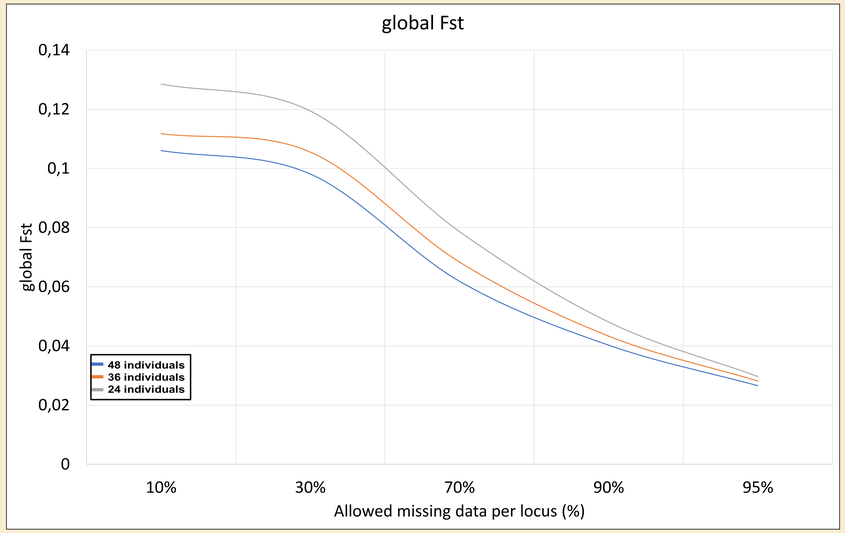
MAIN GOAL

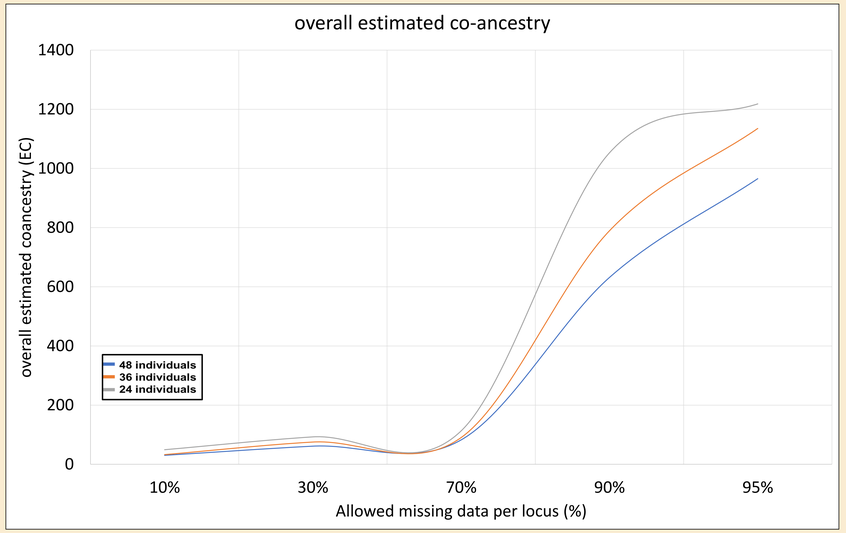
Observations and practical recommendations for maximizing genetic diversity information and reduced missing data on case study of small populations in RAD-seq studies.

MATERIALS AND METHODS

Eokochia saxicola is an endangered species and is also a plant species with extremely small populations (census size 240 individual). From double digest restriction site-associated DNA sequencing (ddRAD-seq) dataset, we test the different strategies to estimated the global Fst, overall Ho (observed heterozygosity), and overall EC (estimated co-ancestry), applying a reduction of the number of SNPs through progressively filtering of missing data (from 10% to 95%) and a reduction the number of examined individuals to increase the number of shared reads (48-36-24 individuals). The missing data was filtered with VCFTOOLS v0.1.14 (Danecek et al., 2011). The estimation of global Fst and overall Ho were calculated with the R package adegenet v2.02 (Jombart, 2008) and hierfstat (Goudet, 2005), while the overall EC was implemented with fineRADstructure software (Malinsky et al., 2018).

DATA FLOW





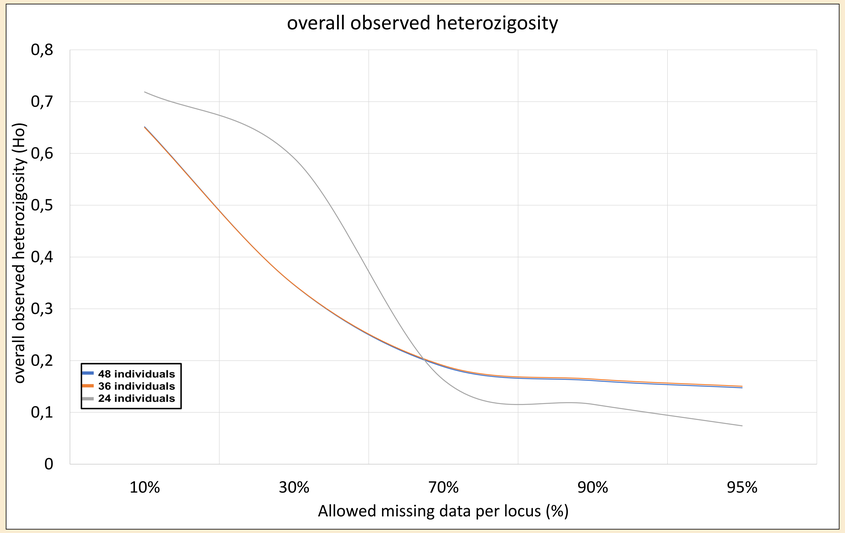


Figure1. The imputation accuracy is plotted against the global Fst (A), overall EC (B), and overall Ho (C) at five levels of missing data (10-30-70-90-95%) for three dataset (48-36-24 individuals).

Independent of the number of individuals, with less rigorous filtering (e.g. loci shared by 10-30% of individuals and higher amount of missing data), the loci are differentiated among diverged populations, generating a higher global Fst. In contrast, when loci with missing data are excluded (i.e. loci shared by 95% of individuals), these are shared among diverged populations and reduce the global Fst value (**Figure1A**). For similar reasons, to reduce the quantity of missing data favouring retained loci (e.g. loci shared by 95 % of individuals) determined an increase in the overall EC (**Figure1B**).

In contrast, the overall Ho value is very high with less stringent filtering (10-30% missing) in the dataset of 24 individuals. For analogous situations, reducing the missing data (i.e. loci shared by 95 % of individuals) determined a decrease in the overall Ho (**Figure1C**).

The different imputation strategies produced a substantial accuracies. Combining imputation strategies can enhance the Fst, EC and Ho estimation on small RAD-seq dataset.

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